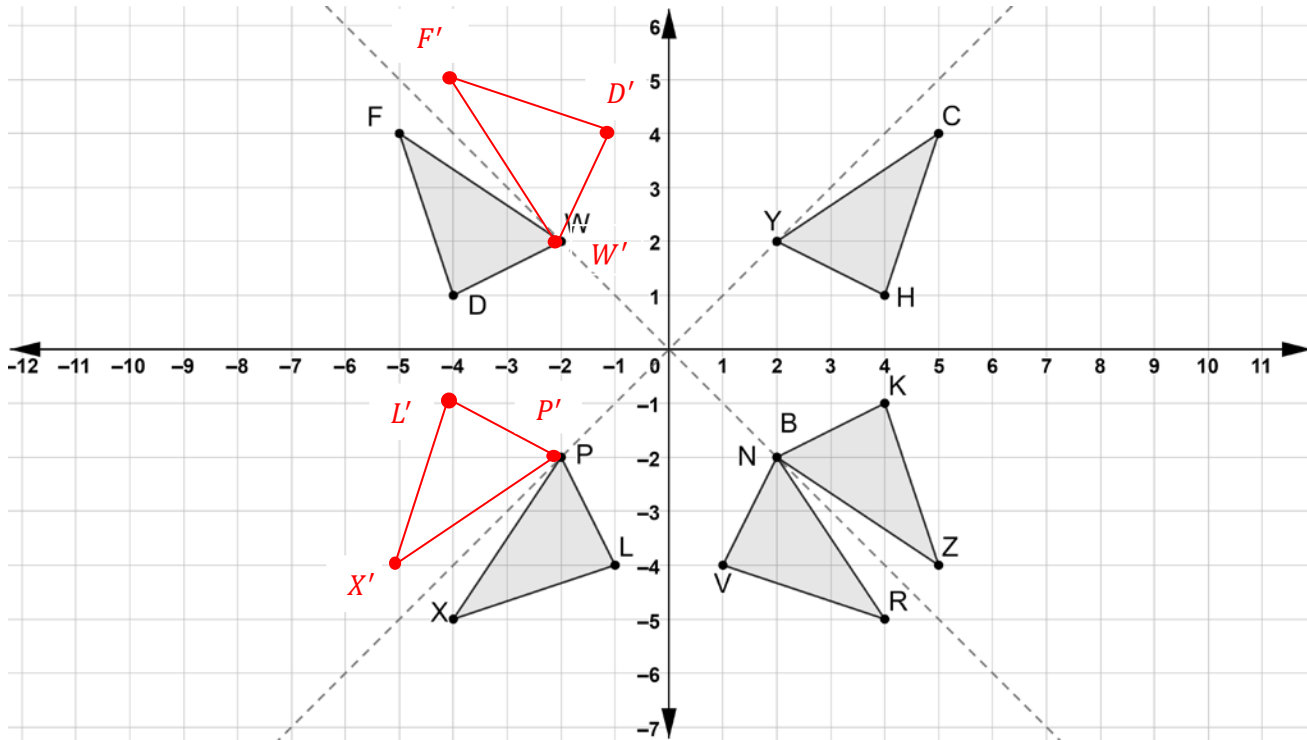


Geometry
Unit 4
Lesson 2 Practice

Name Answer Key
Date _____
Period _____

Six triangles are shown on the coordinate grid.



1. Draw the reflection of $\triangle PLX$ across the line $y = x$. Label the reflection as $\triangle P'L'X'$.
2. Draw the reflection of $\triangle FWD$ across the line $y = -x$. Label the reflection as $\triangle F'W'D'$.
3. Select all of the triangles that are reflections of $\triangle CHY$.
 - ☒ $\triangle FDW$
 - ☒ $\triangle XLP$
 - ☐ $\triangle RVN$
 - ☒ $\triangle ZKB$
 - ☐ $\triangle P'L'X'$
 - ☐ $\triangle F'W'D'$

4. The algebraic description of the reflection of $\triangle BKZ$, the preimage, onto $\triangle NVR$, the image, is $(x, y) \rightarrow (-y, -x)$. Describe how the coordinates of the vertices of $\triangle BKZ$, the preimage, and the coordinates of $\triangle NVR$ the image can be used to determine the line of reflection.

When the coordinates switch places and become opposite, the figure is reflected over the line $y = -x$.

5. Which triangle is the image of reflecting $\triangle FDW$ across the x -axis?

$\triangle X'L'P'$

6. Which triangle is the image of reflecting $\triangle CHY$ across the y -axis?

$\triangle FDW$

7. Describe how the coordinates of the vertices of $\triangle F'W'D'$, the preimage, and the coordinates of $\triangle XPL$ the image can be used to determine the algebraic description of the reflection.

The x -coordinates of $\triangle F'W'D'$ do not change and the y -coordinates become opposite creating the algebraic description $(x, y) \rightarrow (x, -y)$.

8. Describe how the coordinates of the vertices of $\triangle ZKB$, the preimage, and the coordinates of $\triangle CHY$, the image can be used to determine the algebraic description of the reflection.

The x -coordinates of $\triangle ZKB$ do not change and the y -coordinates become opposite creating the algebraic description $(x, y) \rightarrow (x, -y)$.

9. Write the algebraic description for the reflection of $\triangle NVR$, the preimage, onto $\triangle PLX$, the image.

$(x, y) \rightarrow (-x, y)$

10. Write the algebraic description for the reflection of $\triangle P'L'X'$, the preimage, onto, $\triangle BKZ$, the image.

$(x, y) \rightarrow (-x, y)$